

SWEAR: A Multimodal Dataset and Embodied Emotion Simulation System for Understanding Hyperhidrosis through Interactive Wearable Jewellery

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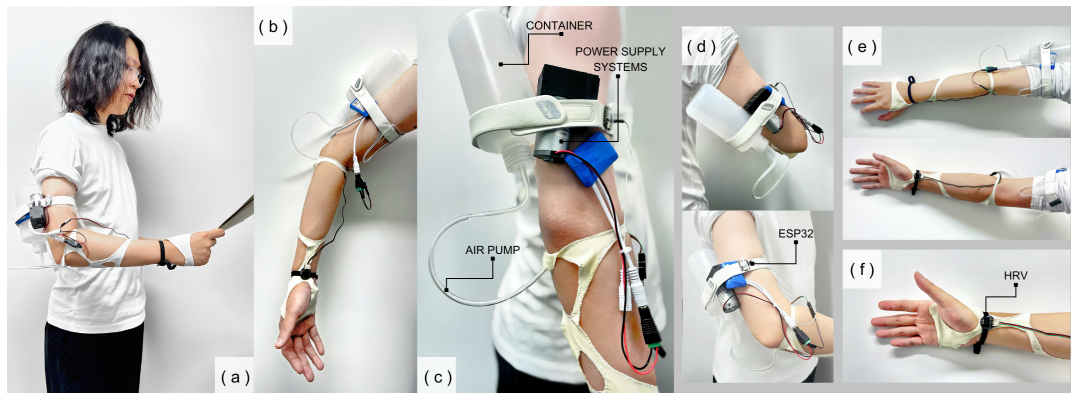


Figure 1: (a) Overall display of device wear; (b) Connection of the air pump to the container; (c) Power supply systems for equipment; (d) Storage for hagfish mucus; (e) Side view of the unit showing the battery module connected to the sensor and the connection of the reservoir battery to the air pump; (f) Heartbeat HRV Sensor Devices.

Abstract

SWEAR (Sweat-induced Emotional Affective Recognition) addresses the underexplored psychosocial impact of palmo/plantar hyperhidrosis in HCI. This wearable jewelry system simulates hyperhidrosis experiences through: Multimodal sensing (PPG/speech affect); Bio-inspired fluidic actuators; Hybrid 1D-CNN/LSTM model with cross-modal attention. Wizard-of-Oz testing with semi-structured interviews and UEQ evaluation demonstrated SWEAR's capacity to elicit reflective emotional responses and enhance empathy toward patients.

*Both authors contributed equally to this research.

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CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; • **Applied computing** → *Media arts*.

Keywords

Hyperhidrosis, Multimodal, Wearable, Affective Computing, Empathy

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1 BACKGROUND AND RESEARCH FRAMEWORK

Hyperhidrosis affects 4.8% of the U.S. population (15.8M) [Doolittle et al. 2016], yet remains underdiagnosed, with only 38% seeking treatment due to stigma. Patients exhibit elevated anxiety/depression rates [Bahar et al. 2016], with 75% avoiding social situations and 33%

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altering careers (International Hyperhidrosis Society, 2023). Drawing on embodied cognition theory, we develop wearable jewellery that simulates hyperhidrosis experiences to bridge this empathy gap. By materializing the invisible burden—social avoidance, career impacts, and mental health consequences—our system challenges stigma and promotes care-seeking.

Affective technology research builds on three pillars: Picard's (1997) emotion-aware systems enabling multimodal inference (PPG/skin conductance/facial-speech analysis) [Koelstra et al. 2011; Soleymani et al. 2017]; Wilson's (2002) embodied cognition theory linking bodily interactions to emotional processing [Wilson 2002]; and Wright and McCarthy's (2004) experience-centered design prioritizing emotional reflection. This foundation informs contemporary wearables with haptic/audio-visual feedback [Yin et al. 2021] and tangible interfaces [Mencarini et al. 2019]. Interactive jewellery thus emerges as dual-purpose artifacts mediating technical and emotional engagement with health experiences.

2 Method

Our interdisciplinary approach integrates affective computing, embodied cognition, and experience-centered design to investigate emotional understanding via wearable technology. We developed an arm-mounted device simulating hyperhidrosis sensations (fig 2), inspired by hagfish mucus defence mechanisms.

The mesh-structured silicone jewellery contains color-coded fluid channels reflecting autonomic states: red (stress-induced high heart rate), nude (baseline), and blue (parasympathetic dominance). These channels guide synthetic mucus flow to replicate physiological sweating patterns.

As shown in fig 1, the system detects emotional states through photoplethysmography (PPG) sensors measuring heart rate variability (HRV) metrics (LF/HF ratio, SDNN, RMSSD), with planned microphone integration for vocal affect analysis. A Raspberry Pi module processes sensor data in real-time. Upon detecting physiological stress (elevated HRV or vocal jitter), pumps release hydrogel from reservoirs onto the skin surface. The fluid release intensity is dynamically modulated through multimodal sensor fusion, creating proportional sweating responses to the user's emotional states.

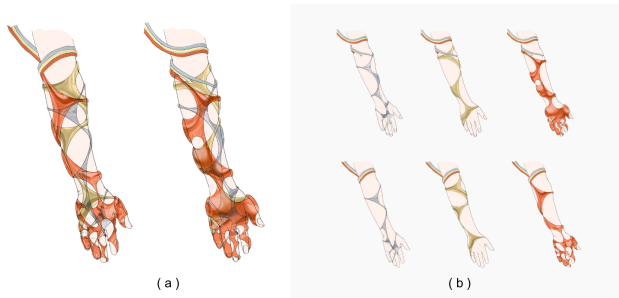


Figure 2: (a) Bionic prototype simulating skin structure; (b) Cross-section diagram

3 DATA COLLECTION AND ANALYSIS

Our multimodal framework captured physiological, behavioural, and contextual data to evaluate the SWEAR system's efficacy in simulating hyperhidrosis and fostering empathy. We employed a Wizard-of-Oz approach with six participants divided into three balanced groups:

- **Patients (P1, P2):** Diagnosed with palmo-plantar hyperhidrosis (student athlete/recreational athlete)
- **Aware Non-Patients (P4, P5):** No condition but contextual familiarity (teacher with affected family/medical doctor)
- **Unfamiliar Participants (P3, P6):** No prior exposure (general teacher/psychotherapist)

Each completed 15–20 minute scenario session (public speaking/social interaction) wearing the SWEAR prototype. The bio-interactive jewellery integrated PPG sensing and fluidic sweat simulation triggered by physiological cues. Physiological stress was quantified through HRV metrics (SDNN, RMSSD, LF/HF) with LF/HF >1.5 indicating sympathetic overactivation ($r = 0.72$, $p < 0.01$). Sweat events were defined by sustained LFnu >60% over 5 minutes concurrent with skin conductance slope > $0.5 \mu S/s$, triggering staged cooling ($\Delta T \geq -2^\circ C$) and breathing feedback (fewer than 12 breaths/minute).

Speech emotion recognition used a ReSpeaker 4-Mic array (>18dB SNR), extracting pitch jitter (DTW), HNR (cepstral), and tempo variation (CTC phoneme alignment). These vocal features were synchronized with HRV metrics as shown in fig 3(a). Real-time multimodal fusion occurred on a Jetson Nano edge node using a lightweight model with:

- 1D-CNN processing MFCC speech features
- LSTM decoding RR intervals
- Cross-modal attention weighting inputs

TensorRT acceleration achieved 45ms latency at 1.2W power, with hardware pulse synchronization ($\pm 0.1ms$).

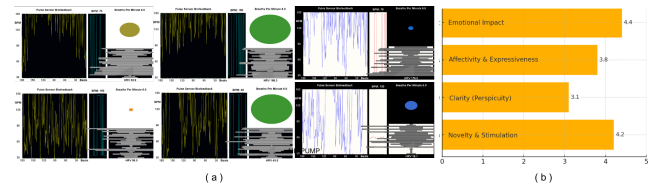


Figure 3: (a) Pulse sensor biofeedback across scenarios; (b) UEQ scores for SWEAR prototype (1=very low, 5=very high)

4 RESULTS AND DISCUSSION

Participant interviews revealed distinct perspectives across groups:

- **Patients (P1, P2)** emphasized social discomfort and exhaustion during physical activity
- **Unfamiliar participants (P3, P6)** initially viewed sweating as minor annoyance
- **Contextually aware non-patients (P4, P5)** reported enhanced emotional understanding

Notably, the physician (P5) stated: "The device clarified when visible symptoms cause emotional shutdown." Practical suggestions

included adding educational interfaces for physiological awareness (proposed by P3, P4).

Quantitative assessment via User Experience Questionnaire (UEQ) in fig 3(b) confirmed these findings:

- Highest scores: Emotional Impact (4.4/5), Novelty & Stimulation (4.2/5)
- Lowest score: Perspicuity (3.1/5)

The strong emotional impact rating (4.4/5) validates the system's capacity to trigger embodied empathy through physiological simulation.

5 CONCLUSION & FUTURE WORK

We present SWEAR (Sweat-induced Emotional Affective Recognition), a multimodal framework combining physiological sensing, fluidic actuation, and cross-modal fusion to enhance hyperhidrosis empathy. Future work employs patient co-design for ergonomic refinement and longitudinal studies assessing cognitive/behavioral impacts. The modular architecture will be publicly released for clinical empathy training, psychoeducational tools, and somaesthetic performance applications, with ethical oversight ensuring patient-centered adaptations.

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